

CONSTANTS	
Parameter	Value
Angle of Attack	5°
Freestream Velocity	175 m/s
Air Density	1.225 kg/m ³
Ideal Power	2,647.6 W
Ideal RPM	112,000 RPM
Ideal Torque	0.22 Nm
Reynolds Number (Re)	7.5 * 10 ⁶
Mass Airplane	1650kg

Airfoil	Configuration	Lift Coefficient (CL)	Lift Penalty	Drag Coefficient (CD)	Drag Penalty	Additional Thrust Needed (W)	Local Velocity (m/s)	Max Power (Betz, W)	Actual Power (W)	Efficiency (% Betz)	RPM	Torque (Nm)
NACA 4412	Baseline (No Turbine)	1.08	-	0.013	-	34061.81	-	-	-	-	-	-
	T-F (Leading Edge)	0.821	23.981	0.049	276.92	168888.86	236	10,958	6,480	59.13	151,700	0.41
	T-M (Mid-Chord)	0.961	11.019	0.04	207.69	117783.56	208	7,502	4,210	56.12	133,700	0.3
	T-B (Trailing Edge)	0.713	33.981	0.089	584.62	353222.65	182	5,030	2,840	56.46	117,000	0.23
	S-F (Internal Front)	1.015	6.019	0.029	123.08	80850.00	125	1,627	453	27.84	80,300	0.05
	S-T (Top Tunnel)	0.94	12.963	0.038	192.31	114394.15	156	3,161	1,303	41.22	100,300	0.12
	S-B (Bottom Tunnel)	0.896	17.037	0.044	238.46	138960.94	164	3,674	1,778	48.39	105,400	0.16

Airfoil	Configuration	Lift Coefficient (CL)	Lift Penalty	Drag Coefficient (CD)	Drag Penalty	Thrust Needed	Local Velocity (m/s)	Max Power (Betz, W)	Actual Power (W)	Efficiency (% Betz)	RPM	Torque (Nm)
CLARK Y	Baseline (No Turbine)	1.02	-	0.015	-	41613.97	-	-	-	-	-	-
	T-F (Leading Edge)	0.765	25.00	0.054	260.00	199747.06	231	10,278	6,023	58.60	148,500	0.39
	T-M (Mid-Chord)	0.908	10.98	0.046	206.67	143357.38	202	6,871	3,841	55.90	129,900	0.28
	T-B (Trailing Edge)	0.663	35.00	0.093	520.00	396933.26	178	4,710	2,652	56.31	114,400	0.22
	S-F (Internal Front)	0.949	6.96	0.033	120.00	98400.16	119	1,405	381	27.12	76,500	0.05
	S-T (Top Tunnel)	0.877	14.02	0.043	186.67	138744.87	152	2,925	1,199	40.99	97,700	0.12
	S-B (Bottom Tunnel)	0.826	19.02	0.049	226.67	167866.53	160	3,413	1,648	48.29	102,900	0.15

Airfoil	Configuration	Lift Coefficient (CL)	Lift Lost (%)	Drag Coefficient (CD)	Drag Gained (%)	Thrust Needed	Local Velocity (m/s)	Max Power (Betz, W)	Actual Power (W)	Efficiency (% Betz)	RPM	Torque (Nm)
NACA 23015	Baseline (No Turbine)	0.81	-	0.011	-	38428.70	-	-	-	-	-	-
	T-F (Leading Edge)	0.616	23.95	0.043	290.91	197531.25	248	12,711	7,881	62.00	159,400	0.47
	T-M (Mid-Chord)	0.729	10.00	0.035	218.18	135859.05	218	8,639	5,028	58.20	140,100	0.34
	T-B (Trailing Edge)	0.527	34.94	0.07	536.36	375868.12	191	5,807	3,252	56.00	122,800	0.25
	S-F (Internal Front)	0.77	4.94	0.025	127.27	91875.00	136	2,098	615	29.31	87,400	0.07
	S-T (Top Tunnel)	0.697	13.95	0.032	190.91	129916.79	167	3,878	1,598	41.21	107,400	0.14
	S-B (Bottom Tunnel)	0.664	18.02	0.037	236.36	157681.85	175	4,465	2,161	48.40	112,500	0.18